Case 03 · Design document: Roll Transparency Log

The step after architecture: classes inside each component, the states a change lives in, who talks to whom and in what order, concurrency, tables, and build order. Characters are fictional.

Four house rules

- **No change without a line.** The officer terminal shows "saved" only after the log returns a receipt with the line number.
- **Nobody holds both number and pseudonym.** The elector number enters the HSM and a pseudonym comes out.
- **The past is not edited.** A correction is a new M-FIX line pointing at the old one. No UPDATE on the leaves table.
- **The public sees district and week.** Booth and exact date appear only in a person's own proof.

Classes by plot

Plot	Class	Does	Does not
Commission	ChangeReader	Reads change rows from the database log	Write to the database
Commission	ReasonValidator	Checks code is one of 11 and evidence attached	Decide if a deletion is right
Commission	PseudonymService	HMAC inside the HSM	Store the number
Commission	EntryBuilder	Canonical line (sorted JSON)	See an elector number
Commission	LogWriter	Appends leaves, signs a head each minute	Change an existing leaf
Commission	TileStore	Tiles of 256 hashes	Delete a tile
Commission	Reconciler	Rebuilds the roll from the log nightly	Fix on its own
Public	CheckService	Number + DOB → lines and proofs	Store the request
Public	ProofVerifier	Rebuilds the head on the phone	Trust the server
Public	Notifier	SMS on the day of change	Send an elector number
Public	AggregateFeed	District/week counts	Publish a booth
Witnesses	Witness	Checks consistency and cosigns	Sign a non-extending head
Tribunals	AppealIntake	Appeal with line id	Accept without a line

Protocol: self-check

`POST /check` with elector number, date of birth and known tree size. Responses: **200** lines and proofs; **204** no changes; **400** missing fields; **409** head not cosigned by three witnesses (the phone refuses to show "verified"); **429** rate-limited.

States of one change

`RECEIVED → VALIDATED → PSEUDONYMIZED → APPENDED → CHECKPOINTED → WITNESSED → NOTIFIED`, plus REJECTED (invalid code or missing evidence; nothing saved), PENDING (log unavailable; nothing saved) and SUPERSEDED (an M-FIX came later; the old line stays).

Sequence: one deletion

Officer → ChangeReader → ReasonValidator → HSM → LogWriter (receipt; only now "saved") → witnesses cosign within a minute → SMS to Priya → her phone fetches and verifies the proof.

Concurrency: two officers, one voter, one second

Each line carries `prev`, the last line number of the same pseudonym, under a unique index on (pseudonym, prev). The second change finds a mismatched `prev`, gets 409, and the officer sees the first change before deciding again. The index is the referee; no locks in code.

Tables: two cabinets, no thread between them

Commission: `roll_changes(change_id, elector_no, booth, action, reason, evidence_sha256, officer_id, at)`, `outbox(change_id, status, attempts)`.

Log (public): `leaves(seq, leaf_hash, entry_json)`, `checkpoints(size, root, sig, at)`, `witness_sigs(size, witness_id, sig)`, `aggregates(district, week, reason, count)`.

No shared column identifies a person; joining requires the HMAC key, which never leaves the HSM.

Build order

Version	Built	Exit test
V0.1	Tree, tiles, signed head	A line enters, the head changes, the signature verifies
V0.2	Change reader, codes, evidence	A deletion reaches the log with a code; an invalid code is rejected
V0.3	HSM and pseudonyms	No elector number in any public table
V0.4	One witness, consistency proofs	A rewrite is caught within a minute
V0.5	Self-check, SMS, kiosk	Priya gets "verified"; an uncosigned head shows a warning
V0.6	Reconciler, five witnesses, aggregate feed	A bypassing change is caught; no booth in the feed